NUMERICAL COMPUTATION OF HADAMARD FINITE PART OF SOME HYPERSINGULAR INTEGRALS IN 1, 2 AND 3 DIMENSIONS

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ABSTRACT

Data mining is a good way to find the relationship between raw data and predict the target we want which is also widely used in different field nowadays. In this project, we implement a lots of technology and method in data mining to predict the sale of an item based on its previous sale. We create a strong model to predict the sales. After evaluating this model, we conclude that this model can be used in normal life for future sale's prediction.

1 Introduction

Our project is a competition on Kaggle (Predict Future Sales). We are provided with daily historical sales data (including each products' sale date, block ,shop price and amount). And we will use it to forecast the total amount of each product sold next month. Because of the list of shops and products slightly changes every month. We need to create a robust model that can handle such situations.

2 Notations

For every physical quantity, the corresponding reference space quantity is denoted by a hat, so that, for example, if $\mathbf{x}$ is a point on a d-dimensional variety, then $\hat{\mathbf{x}}$ is the corresponding point in the reference space. If u is a function defined on the variety, then $\hat{u}$ is the corresponding function in the reference space. Furthermore, we denote by $⨍$ the Hadamard finite part of an integral.

3 Model presentation

Let's consider an finite part integral of the form

$$I = ⨍_{\hat{\tau}} f(\hat{\mathbf{y}}) \mathrm{d}\hat{\mathbf{y}}, \qquad f(\hat{\mathbf{y}}) \underset{\hat{y} \to \hat{x}}{\sim} \frac{1}{||\hat{\mathbf{y}} - \hat{\mathbf{x}}||^{d+1}}. \tag{1}$$

4 Injecting Richardson extrapolation into Guiggiani's final formula

4.1 Richardson extrapolation - general idea

For any angle θ, let's consider a finite sequence of n radial points $\rho_i(\theta)$ such that

$$0 < \rho_1(\theta) < \rho_2(\theta) < \cdots < \rho_n(\theta) < \hat{\rho}(\theta), \tag{2}$$

Recall: we can express the two first terms of the Laurent expansion of the complete polar kernel $F(\rho, \theta) = \rho K(\chi(\hat{\mathbf{x}}), \chi(\hat{\mathbf{y}}(\rho, \theta)))J(\hat{\mathbf{y}}(\rho, \theta))\hat{u}(\hat{\mathbf{y}}(\rho, \theta))$ as the following limits

$$F_{-2}(\theta) = \lim_{\rho\to 0} \rho^2 F(\rho, \theta), \qquad F_{-1}(\theta) = \lim_{\rho\to 0} \rho \left(F(\rho, \theta) - \frac{F_{-2}(\theta)}{\rho^2} \right). \tag{3}$$

such that we can write the complete polar kernel as

$$F(\rho, \theta) = \frac{F_{-2}(\theta)}{\rho^2} + \frac{F_{-1}(\theta)}{\rho} + \underset{\rho\to 1}{O}(1) \tag{4}$$

Let $c_{1,i}$ and $c_{2,i}$ be the coefficients used in the Richardson extrapolation of order n to compute the previous limits, so that we can write

$$\boxed{F_{-2}(\theta) \approx \widetilde{F}_{-2}(\theta) = \sum_{i=1}^{n} c_{2,i}\rho_i^2(\theta)F(\rho_i(\theta), \theta)}, \qquad F_{-1}(\theta) \approx \widetilde{F}_{-1}(\theta) = \sum_{i=1}^{n} c_{1,i}\rho_i(\theta) \left(F(\rho_i(\theta), \theta) - \frac{\widetilde{F}_{-2}(\theta)}{\rho_i^2(\theta)} \right). \tag{5}$$

By injecting the formula of $\widetilde{F}_{-2}(\theta)$ into the formula of $\widetilde{F}_{-1}(\theta)$, so that we can write $\widetilde{F}_{-1}(\theta)$ only as a linear combination of the complete polar kernel evaluated at the radial points $\rho_i(\theta)$, we obtain

$$\widetilde{F}_{-1}(\theta) = \sum_{i=1}^{n} c_{1,i}\rho_i(\theta)F(\rho_i(\theta), \theta) - \sum_{i=1}^{n} c_{1,i}\frac{1}{\rho_i(\theta)} \sum_{j=1}^{n} c_{2,j}\rho_j^2(\theta)F(\rho_j(\theta), \theta).$$

By letting $P_n(\theta) = \sum_{k=1}^{n} \frac{1}{\rho_k(\theta)}$ be a constant that is a polynomial in $\cos\theta$ and $\sin\theta$, we can write the previous formula as

$$\boxed{\widetilde{F}_{-1}(\theta) = \sum_{i=1}^{n} F(\rho_i(\theta), \theta)\rho_i(\theta)\Big(c_{1,i} - c_{2,i}P_n(\theta)\rho_i(\theta)\Big)} \tag{6}$$

4.2 Lagrange basis method

For this section and the following, we will no longer use θ in the notation, it will be implicit. Let's denote the regular function $G(\rho) = \rho^2 F(\rho)$, so that $G(0) = F_{-2}$ and $G'(0) = F_{-1}$. Let $\ell_i(\rho)$ be the Lagrange basis of degree $n-1$ associated to the radial points ρ_i, defined as

$$\ell_i(\rho) = \prod_{j\neq i} \frac{\rho - \rho_j}{\rho_i - \rho_j}, \qquad i = 1, \ldots, n. \tag{7}$$

The lagrange interpolant of G and its derivative are given by

$$\Pi_{n-1}G(\rho) = \sum_{i=1}^{n} G(\rho_i)\ell_i(\rho), \qquad (\Pi_{n-1}G)'(\rho) = \sum_{i=1}^{n} G(\rho_i)\ell_i'(\rho). \tag{8}$$

So that the numerical Laurent's coefficients, denoted $\widetilde{F}_{-2}$ and $\widetilde{F}_{-1}$, are simply the evaluation of the interpolant and its derivative at the origin $\rho = 0$:

$$\widetilde{F}_{-2} = \Pi_{n-1}G(0) = \sum_{i=1}^{n} G(\rho_i)\ell_i(0), \qquad \widetilde{F}_{-1} = (\Pi_{n-1}G)'(0) = \sum_{i=1}^{n} G(\rho_i)\ell_i'(0). \tag{9}$$

with $\ell_i(0) = \prod_{j\neq i} \frac{\rho_j}{\rho_j - \rho_i}$ and $\ell_i'(0) = \ell_i(0)\left(\frac{1}{\rho_i} - P_n\right)$, so that we can finally write the Laurent's coefficients as

$$\widetilde{F}_{-2} = \sum_{i=1}^{n} \ell_i(0)\rho_i^2 F(\rho_i), \qquad \widetilde{F}_{-1} = \sum_{i=1}^{n} \ell_i(0)\rho_i\left(1 - P_n\rho_i\right)F(\rho_i). \tag{10}$$

This is a particular case of the Richardson extrapolation, with the two families of coefficients $c_{1,i}$ and $c_{2,i}$ collapsing into a single family of coefficients $\ell_i(0)$:

$$c_{1,i} = c_{2,i} = \ell_i(0) = c_i = \prod_{j\neq i} \frac{\rho_j}{\rho_j - \rho_i}.$$

Finally, the numerical Laurent's coefficients can be written as

$$\widetilde{F}_{-2}(\theta) = \sum_{i=1}^{n} A_i(\theta)F(\rho_i(\theta),\theta), \qquad \widetilde{F}_{-1}(\theta) = \sum_{i=1}^{n} B_i(\theta)F(\rho_i(\theta),\theta). \tag{11}$$

With $A_i(\theta) = c_i(\theta)\rho_i^2(\theta)$ and $B_i(\theta) = c_i(\theta)\rho_i(\theta)\left(1 - P_n(\theta)\rho_i(\theta)\right)$.

Each of these formula simply consist in the numerical evaluation of the polar kernel at the radial points $\rho_i(\theta)$, multiplied by some weights.

4.3 Final quadrature rule

Let's then denote $I(\theta)$ the integral of the complete polar kernel over the radial direction, defined as:

$$I(\theta) = \int_0^{\hat{\rho}(\theta)} \left\{F(\rho,\theta) - \frac{F_{-1}(\theta)}{\rho} - \frac{F_{-2}(\theta)}{\rho^2}\right\} \mathrm{d}\rho + F_{-1}(\theta)\ln\left|\frac{\hat{\rho}(\theta)}{\beta(\theta)}\right| - F_{-2}(\theta)\left\{\frac{\gamma(\theta)}{\beta^2(\theta)} + \frac{1}{\hat{\rho}(\theta)}\right\} \tag{12}$$

A final quadrature rule for it is the following:

$$I(\theta) \approx \widetilde{I}(\theta) = \sum_{i=1}^{n} \widetilde{w}_i(\theta)F(\rho_i(\theta),\theta) \tag{13}$$

With the new weights $\widetilde{w}_i(\theta)$ defined as

$$\widetilde{w}_i(\theta) = w_i(\theta) - A_i(\theta)\left\{W_{-2}(\theta) + \frac{\gamma(\theta)}{\beta^2(\theta)} + \frac{1}{\hat{\rho}(\theta)}\right\} + B_i(\theta)\left\{\ln\left|\frac{\hat{\rho}(\theta)}{\beta(\theta)}\right| - W_{-1}\right\}$$

and the auxiliary constants :

- $w_i(\theta) = \frac{\hat{\rho}(\theta)}{2} w_i^{\mathrm{Ref}}$ are the weights of the reference quadrature over the segment $[-1, 1]$.
- $W_{-1} = \sum_{i=1}^{n} \frac{w_i(\theta)}{\rho_i(\theta)} = \sum_{i=1}^{n} \frac{w_i^{\mathrm{Ref}}}{1+x_i^{\mathrm{Ref}}}$.
- $W_{-2}(\theta) = \sum_{i=1}^{n} \frac{w_i(\theta)}{\rho_i^2(\theta)} = \frac{2}{\hat{\rho}(\theta)} \sum_{i=1}^{n} \frac{w_i^{\mathrm{Ref}}}{\left(1+x_i^{\mathrm{Ref}}\right)^2}$.

Remark: in the case of Gauss-Legendre quadrature, the constants W_{-1} and W_{-2} can be computed analytically as:

$$W_{-1} = 2H_n, \qquad W_{-2}(\theta) = \frac{2}{\hat{\rho}(\theta)} n(n+1) \tag{14}$$

With $H_n = \sum_{i=1}^{n} \frac{1}{i}$ the n-th harmonic number. These last two formulas are, for the moment, a conjecture. (Elaborate to prove it?)

4.4 Convergence and error study

To begin our reasoning, assume that we have computed the Laurent's coefficients up to a relative error ϵ_{-k}, $k \in \{1, 2\}$:

$$F_k = \widetilde{F}_{-k}(1 + \epsilon_{-k}), \qquad k \in \{1, 2\} \tag{15}$$

Now let's denote

$$G(\rho, \theta) = F(\rho, \theta) - \frac{F_{-1}(\theta)}{\rho} - \frac{F_{-2}(\theta)}{\rho^2} \tag{16}$$

$$\widetilde{G}(\rho, \theta) = F(\rho, \theta) - \frac{\widetilde{F}_{-1}(\theta)}{\rho} - \frac{\widetilde{F}_{-2}(\theta)}{\rho^2} \tag{17}$$

$$\tag{18}$$

$$I = \int_0^{2\pi} \int_0^{\hat{\rho}(\theta)} G(\rho, \theta) \mathrm{d}\rho \mathrm{d}\theta, \qquad \widetilde{I} = \int_0^{2\pi} \int_0^{\hat{\rho}(\theta)} \widetilde{G}(\rho, \theta) \mathrm{d}\rho \mathrm{d}\theta \tag{19}$$

On one hand, after the theory of the Bernstein's ellipses, if we assume that G and $\widetilde{G}$ are analytic, we can make a error majoration regarding the Gauss-Legendre quadrature of I and $\widetilde{I}$:

$$|I - Q_n[G]| \leq \frac{64}{15} \frac{M}{(1 - \rho_B^{-2})\rho_B^{2n}} \tag{20}$$

$$\left|\widetilde{I} - Q_n[\widetilde{G}]\right| \leq \frac{64}{15} \frac{\widetilde{M}}{(1 - \widetilde{\rho}_B^{-2})\widetilde{\rho}_B^{2n}} \tag{21}$$

On the other hand, we can substract $\widetilde{I}$ to I to get:

$$|I - \widetilde{I}| \leq C(\hat{x}) \sum_{k=1}^{2} \frac{\epsilon_{-k}}{\rho_{\min}^k} (1 + \max F_{-k}) \tag{22}$$

where $C(\hat{x}) = \displaystyle\int_0^{2\pi} \hat{\rho}(\theta) \mathrm{d}\theta$.

Thus,

$$|I - Q_n[\widetilde{G}]| = |I - \widetilde{I} + \widetilde{I} - Q_n[\widetilde{G}]| \quad \leq |I - \widetilde{I}| + |\widetilde{I} - Q_n[\widetilde{G}]| \leq C(\hat{x}) \sum_{k=1}^{2} \frac{\epsilon_{-k}}{\rho_{\min}^k} (1 + \max F_{-k}) + \frac{64}{15} \frac{\widetilde{M}}{(1 - \widetilde{\rho}_B^{-2})\widetilde{\rho}_B^{2n}} \tag{23}$$